

- Description

This module manages a smart oven system, ensuring that:

1. The oven continuously monitors the current cooking temperature and compares it with a target temperature.
2. If the current temperature deviates from the target, the system automatically adjusts the cooking timer.
3. The adjustment is based on numerical calculations: if the temperature is too low, the timer is extended; if too high, the timer is shortened.

This system helps optimize cooking performance by maintaining ideal temperature conditions and adjusting cooking times accordingly.

- Specification

module SmartOvenControl

var currentTemperature, targetTemperature, baseCookingTime,
calculatedCookingTime: real

process updateTemperature(changeAmount: real) returns (calculatedCookingTime: real, currentTemperature: real)

functional scenario form:

currentTemperature < targetTemperature && calculatedCookingTime =
baseCookingTime * 1.1, Cooking time extended due to low temperature.

||

currentTemperature > targetTemperature && calculatedCookingTime =
baseCookingTime * 0.9, Cooking time reduced due to high temperature.

||

currentTemperature = targetTemperature && calculatedCookingTime =
baseCookingTime, Cooking time remains at base value.

end process

end module

- Java Code

```
package code;
class SmartOven {
    private double targetTemperature;
    private double currentTemperature;
    private double baseCookingTime;
    private double calculatedCookingTime;
    public SmartOven(double targetTemperature, double initialTemperature, double baseCookingTime) {
        this.targetTemperature = targetTemperature;
        this.currentTemperature = initialTemperature;
        this.baseCookingTime = baseCookingTime;
        this.calculatedCookingTime = baseCookingTime;
    }
}
```

```

        public void updateTemperature(double changeAmount) {
            currentTemperature += changeAmount;
            adjustCookingTime();
            System.out.printf("Current Temperature: %.2f°C, Calculated Cooking Time: %.2f minutes\n",
                               currentTemperature, calculatedCookingTime);
            if (currentTemperature < targetTemperature - 5) {
                System.out.println("ALERT: Temperature too low! Extending cooking time.");
            } else if (currentTemperature > targetTemperature + 5) {
                System.out.println("ALERT: Temperature too high! Reducing cooking time.");
            } else {
                System.out.println("Temperature is within optimal range.");
            }
        }
        private void adjustCookingTime() {
            if (currentTemperature < targetTemperature) {
                calculatedCookingTime = baseCookingTime * 1.1;
            } else if (currentTemperature > targetTemperature) {
                calculatedCookingTime = baseCookingTime * 0.9;
            } else {
                calculatedCookingTime = baseCookingTime;
            }
        }
        public double getCalculatedCookingTime() {
            return calculatedCookingTime;
        }

        public double getCurrentTemperature() {
            return currentTemperature;
        }
    }
    public class c75_SmartOvenSystem {
        public static void main(String[] args) {
            SmartOven oven = new SmartOven(180.0, 170.0, 30.0);

            // testcase-VT:
            // oven.updateTemperature(5.0);
            // oven.updateTemperature(-10.0);
            // oven.updateTemperature(20.0);
            // oven.updateTemperature(200.0);
            // oven.updateTemperature(-205.0);
            // testcase-FT:

```

```
//      oven.updateTemperature(1.6349678233795336E296);
//      oven.updateTemperature(-1.7976931348623157E308);
//      oven.updateTemperature(0.0);
//      oven.updateTemperature(-2.2250738585072014E-308);
//      oven.updateTemperature(6.365249299686023E293);

//      // testcase-Z3
//      oven.updateTemperature(5.0);
//      oven.updateTemperature(-10.0);
//      oven.updateTemperature(20.0);
//      oven.updateTemperature(200.0);
//      oven.updateTemperature(-205.0);

//      // testcase-UVT:
oven.updateTemperature(5.0);
oven.updateTemperature(-10.0);
oven.updateTemperature(20.0);
oven.updateTemperature(200.0);
oven.updateTemperature(-205.0);
    }
}
```

- Coverage report

1. Test case number: 5
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	1.0 %	8	834	842
✓ src	1.0 %	8	834	842
✓ code	1.0 %	8	834	842
✓ c75_SmartOvenSystem.java	100.0 %	8	0	8
> c75_SmartOvenSystem		0	0	0
> SmartOven	100.0 %	8	0	8

3. Vibration testing instruction coverage: 93.2%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.6 %	123	20,885	21,008
✓ src	0.6 %	123	20,885	21,008
✓ code	0.6 %	123	20,885	21,008
✓ c75_SmartOvenSystem.java	93.2 %	123	9	132
> c75_SmartOvenSystem	88.5 %	23	3	26
> SmartOven	94.3 %	100	6	106

4. Fuzz testing branch coverage: 75%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.7 %	6	836	842
✓ src	0.7 %	6	836	842
✓ code	0.7 %	6	836	842
✓ c75_SmartOvenSystem.java	75.0 %	6	2	8
> c75_SmartOvenSystem		0	0	0
> SmartOven	75.0 %	6	2	8

5. Fuzz testing instruction coverage: 87.9%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.6 %	116	20,892	21,008
▼ src	0.6 %	116	20,892	21,008
▼ code	0.6 %	116	20,892	21,008
▼ c75_SmartOvenSystem.java	87.9 %	116	16	132
> c75_SmartOvenSystem	88.5 %	23	3	26
> SmartOven	87.7 %	93	13	106

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c75_SmartOvenSystem.java	100.0 %	8	0	8
> c75_SmartOvenSystem		0	0	0
> SmartOven	100.0 %	8	0	8

7. Z3 instruction coverage: 93.2%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.6 %	123	20,885	21,008
▼ src	0.6 %	123	20,885	21,008
▼ code	0.6 %	123	20,885	21,008
▼ c75_SmartOvenSystem.java	93.2 %	123	9	132
> c75_SmartOvenSystem	88.5 %	23	3	26
> SmartOven	94.3 %	100	6	106

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c75_SmartOvenSystem.java	100.0 %	8	0	8
> c75_SmartOvenSystem		0	0	0
> SmartOven	100.0 %	8	0	8

9. Unimproved V-method instruction coverage: 93.2%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.6 %	123	20,885	21,008
▼ src	0.6 %	123	20,885	21,008
▼ code	0.6 %	123	20,885	21,008
▼ c75_SmartOvenSystem.java	93.2 %	123	9	132
> c75_SmartOvenSystem	88.5 %	23	3	26
> SmartOven	94.3 %	100	6	106